

```

# -----
# 1. Import Libraries
# -----
import pandas as pd
from sklearn.model_selection import train_test_split
from sklearn.ensemble import RandomForestClassifier
from sklearn.metrics import accuracy_score, precision_score, recall_score, f1_score

# -----
# 2. Load Dataset
# -----
df = pd.read_csv('/content/OnlineRetail.csv', encoding='latin1')

# -----
# 3. Data Cleaning
# -----
df = df.dropna(subset=['CustomerID'])
df = df[(df['Quantity'] > 0) & (df['UnitPrice'] > 0)]

df['TotalPrice'] = df['Quantity'] * df['UnitPrice']

# -----
# 4. Customer Aggregation
# -----
customer_df = df.groupby('CustomerID').agg({
    'TotalPrice': 'sum',
    'InvoiceNo': 'nunique',
    'Quantity': 'sum'
})

customer_df.columns = ['TotalSpent', 'Transactions', 'Quantity']

# -----
# 5. Target Variable (Segmentation)
# -----
median_value = customer_df['TotalSpent'].median()
customer_df['Segment'] = (customer_df['TotalSpent'] > median_value).astype(int)

# -----
# 6. Features (NO DATA LEAKAGE)
# -----
X = customer_df[['Transactions', 'Quantity']] # Removed TotalSpent
y = customer_df['Segment']

# -----
# 7. Train-Test Split
# -----
X_train, X_test, y_train, y_test = train_test_split(
    X, y, test_size=0.2, random_state=42
)

# -----
# 8. Model (Improved Random Forest)
# -----
model = RandomForestClassifier(
    n_estimators=150,
    max_depth=6,
    min_samples_split=5,
    random_state=42
)

model.fit(X_train, y_train)

# -----
# 9. Prediction
# -----
y_pred = model.predict(X_test)

# -----
# 10. Evaluation
# -----
print("Accuracy:", accuracy_score(y_test, y_pred))
print("Precision:", precision_score(y_test, y_pred))
print("Recall:", recall_score(y_test, y_pred))
print("F1 Score:", f1_score(y_test, y_pred))

```

```
Accuracy: 0.9182027649769585  
Precision: 0.9339622641509434  
Recall: 0.9020501138952164  
F1 Score: 0.9177288528389339
```